

Department of Information Technology

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch : IV Semester-B. Tech Information Technology

Course Code & Title : IT3401 Web Essentials

Name of the Faculty member : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic : Unit 1 / Domain Name**
- **Course Outcome : CO1**
- **Topic Learning Outcome: TLO6**
- **Activity Chosen : Mind Map**
- **Justification:**
 - A mind map activity is a creative visual approach that allows students to organize and present concepts systematically. It encourages learners to identify relationships among ideas, which enhances understanding, critical thinking, and long-term retention of concepts.
 - This activity was chosen to improve students' understanding of the Domain Name System (DNS) and the communication process involved when accessing websites through domain names.
 - The mind map activity will help students understand the concept of domain names and the role of DNS servers, IP addresses, web browsers, and web servers in web communication.
- **Time Allotted for the Activity: 15 Minutes**
- **Details of the Implementation:**
 - The instructor explained the concept of domain names, the role of DNS servers, and how a web browser communicates with a web server through the Domain Name System (DNS).
 - The instructor also discussed the process of DNS resolution where a domain name is translated into an IP address before establishing communication with the web server.
 - After clarifying students' doubts, the faculty instructed the students to draw a mind map related to the Domain Name communication process within 15 minutes.
 - Students visually represented the interaction between DNS servers, web browsers, and web servers through mind maps as shown in Fig 2. These sheets were subsequently collected by the instructor.
 - The activity helped students develop a better conceptual understanding of domain names and DNS functionality in web communication.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	3	2	2	2	1	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO 3	PO9	PO10	PSO3
	3	2	2	2	1	3
Justification for correlation	Students will be able to understand the fundamental concepts of Domain Names, DNS resolution, and browser-server communication.	Students will be able to analyze how domain names are translated into IP addresses and how requests are routed through DNS servers.	Students visually designed and represented the communication process using mind maps.	Students enhanced collaborative learning and discussion skills while interacting with peers during the activity.	Students will be able to communicate technical concepts clearly through organized visual representation.	The knowledge gained helped students understand real-world web communication and internet technologies.

• Images / Screenshot of the practice:



Fig 1: Mind Map Activity

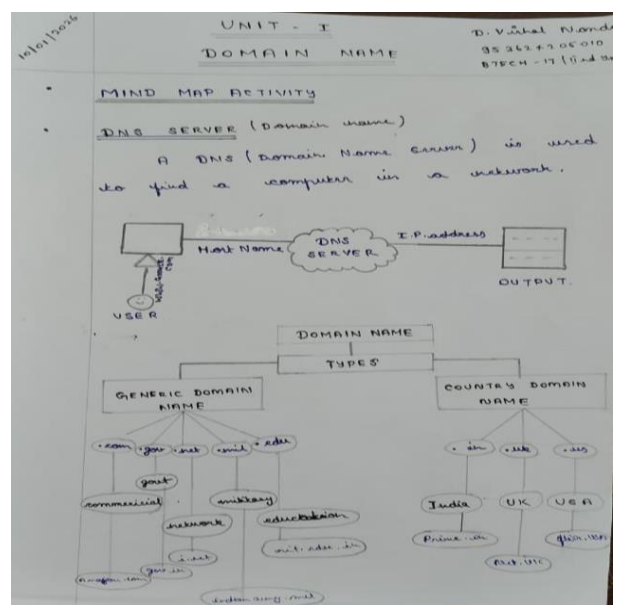


Fig 2: Sample Sheet of Mind Map Activity of Student D.Vishal Nanda

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported that the mind map activity made it easier for them to understand the role of DNS servers in establishing communication between web browsers and web servers.
- The visual representation helped students remember important concepts such as domain names, IP address resolution, DNS lookup, and HTTP communication more effectively.
- The activity also encouraged active peer interaction, enabling students to discuss concepts freely and clarify their doubts collaboratively.

- ❖ ***Benefit of the practice:***

- The mind map activity created an engaging learning environment that improved students involvement and understanding of the concepts.
- The activity assisted students in organizing and understanding the relationship between domain names and DNS processes.
- The activity encouraged students to think visually and logically about how internet communication occurs.
- Through this activity, Students developed better clarity regarding the interaction between web browsers, DNS servers, and web servers.

- ❖ ***Challenges faced in implementation:***

- A few students faced difficulty in clearly depicting the interaction among DNS servers, browsers, and web servers within the domain name resolution process.
- Certain students required additional guidance to clearly represent the process of domain name resolution and HTTP communication.

In the following class, I guided the students on how to clearly illustrate the domain name resolution process by connecting the roles of DNS servers, IP addresses, web browsers, and web servers, and also instructed them to present their mind maps in a well-organized manner.

References:

- https://www.ritrjpm.ac.in/images/computer-science/5_CS8591_Mindmap.pdf

Signature of Faculty Member

HOD